Anterior cingulate cortex · hierarchical baseline DETECTION

New-section p:95% predictive
Anatomy μ:95% credible
Posterior μ mean
Measured pooled fraction

A Fosb

ACAd1 (n=2,668; S=17)
ACAd2/3 (n=1,592; S=16)
ACAd5 (n=1,928; S=16)
ACAd6a (n=1,543; S=15)
ACAd6b (n=53; S=10)
ACAv1 (n=2,047; S=11)
ACAv2/3 (n=1,872; S=11)
ACAv5 (n=1,848; S=11)
ACAv6a (n=897; S=11)
ACAv6b (n=134; S=10)

0.0 0.2 0.4 0.6 0.8 1.0
Detection probability

B Egr3

ACAd1 (n=2,668; S=17)
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0.0 0.2 0.4 0.6 0.8 1.0
Detection probability

C S1pr1

ACAd1 (n=2,668; S=17)
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0.0 0.2 0.4 0.6 0.8 1.0
Detection probability

D New section, actual n

Fosb
Egr3
S1pr1
C730002L08Rik
Fosl2
Col19a1
Pcdh8
Iqgap2
Usp43
Rorb
Nominal95%

0.00 0.25 0.50 0.75 1.00
Held-out count PI coverage

E Held-out fraction error

Fosb
Egr3
S1pr1
C730002L08Rik
Fosl2
Col19a1
Pcdh8
Iqgap2
Usp43
Rorb

0.00 0.05 0.10 0.15
Fraction MAE
(section × anatomy)

F Grid sensitivity
full-data μ means

Fosb
Egr3
S1pr1
C730002L08Rik
Fosl2
Col19a1
Pcdh8
Iqgap2
Usp43
Rorb

0.000 0.004 0.008
Maximum |Δ posterior μ mean|

Anatomy Beta-Binomial
Regional training-only baseline
64×48 vs 32×24
64×48 vs 96×64

One unexposed donor; physical sections model heterogeneity, not independent animals. All measured zeros retained.
Fixed μ~Beta(½,½), κ~LogNormal(log20,1.5);64×48 prior-quantile grid;8,192 seeded mixture draws. OOF prior-only groups=0/1280. No PM2.5 effect or new spatial resolution.